American University of Beirut
Department of Computer Science

CMPS 355 — Bioinformatics Algorithms
Research Proposal

Gene-Level vs. Pathway-Level Transcriptomic Representations for Cross-Cohort Classification of Major Depressive Disorder

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1. Introduction

1.1 Background of Study

Major depressive disorder (MDD) is diagnosed primarily through clinical symptoms and behavioral assessment, with no objective biomarker currently established for routine diagnosis (Ang et al., 2025). This has motivated interest in blood transcriptomics, which measures gene-expression patterns in peripheral blood and may provide molecular signals associated with MDD.

Studies have identified reproducible expression differences in MDD, particularly in genes related to immune function (Leday et al., 2018; Wittenberg, Greene, Vértes, Drevets, & Bullmore, 2020).

1.2 Problem Statement and Significance

Although blood transcriptomic differences have been identified in MDD, predictive models based on these signals may not generalize across independent cohorts. Biological heterogeneity, blood-cell composition, laboratory platforms, and processing can introduce cross-dataset variability (Wittenberg et al., 2020).

Limited reproducibility of individual gene-level signatures motivates investigating whether broader pathway-level representations provide more robust cross-cohort features (Gomez Rueda & Bustillo, 2022).

1.3 Research Objectives

The objective is to compare gene-level and pathway-level blood transcriptomic representations and determine whether pathway-level features improve cross-cohort MDD classification.

1.4 Research Question

This research seeks to answer the following question:

> *Do pathway-level transcriptomic representations improve cross-cohort generalization in MDD classification compared with gene-level representations?*

2. Literature Review

2.1 Gene-Level Transcriptomic Classification

Blood gene-expression features have shown potential for MDD classification. Liu et al. (2022) developed a six-gene predictive model, while Bouzid et al. (2023) identified potential transcriptomic biomarkers using machine-learning approaches. However, performance within individual studies does not establish that gene-level signatures will generalize reliably across independent cohorts.

2.2 Pathway-Level Representations

Pathway-based representations aggregate related genes into broader biological signals that may be more stable across datasets. Su, Yoon, and Dougherty (2009) found greater cross-dataset reproducibility for pathway-based classifiers than for individual gene-based classifiers. In MDD, pathway- and gene-group-level analyses have also identified biologically meaningful patterns (Tang, Wu, Dai, Liu, & Liu, 2025; Wang et al., 2025). However, whether pathway-level features improve cross-cohort MDD classification compared with individual genes remains unclear.

2.3 Research Gap

It remains unclear whether pathway-level features provide more robust cross-cohort MDD classification than individual gene-expression features.

3. Methodology

3.1 Research Design and Proposed Method

This study will compare gene-level and pathway-level representations under the same cross-cohort MDD classification framework.

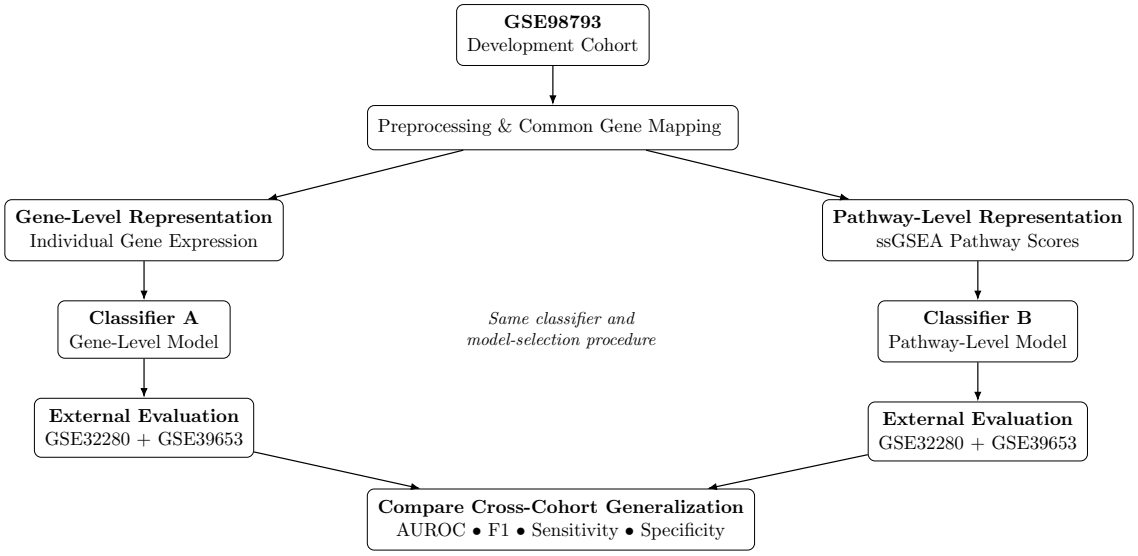


Figure 3.1: Proposed experimental framework for comparison

3.2 Dataset / Data Collection

Publicly available blood transcriptomic datasets will be obtained from the NCBI Gene Expression Omnibus (GEO). GSE98793 will be used for model development, with GSE32280 and GSE39653 reserved for external validation.

Table 3.1: Blood transcriptomic datasets included in the study.

Cohort	MDD	Controls	Sample	Platform	Role
GSE98793	128	64	Whole blood	Affymetrix U133 Plus 2.0	Development
GSE32280	8*	8*	Peripheral blood lymphocytes	Affymetrix U133 Plus 2.0	External validation
GSE39653	21	24	PBMC	Illumina HT-12 v4	External validation

*Pretreatment MDD and corresponding healthy-control samples are used from GSE32280.

GSE32280 provides external validation on the same microarray platform as the development cohort, while GSE39653 provides a stronger validation setting across a different platform and blood sample type.

3.3 Experimental Design and Evaluation

Expression data will first be mapped to common gene identifiers across cohorts. Two feature representations will then be constructed. The gene-level representation will use the top 50 genes ranked using only the training data, while the pathway-level representation will use single-sample Gene Set Enrichment Analysis (ssGSEA) (Barbie et al., 2009) to generate activity scores for the 50 MSigDB Hallmark gene sets (Liberzon et al., 2015). Matching the number of features helps control for dimensionality when comparing the two representations.

Separate gene-level and pathway-level models will be trained on GSE98793 using elastic-net logistic regression (Zou & Hastie, 2005). Feature selection and hyperparameter tuning will be performed within cross-validation using only the development cohort to prevent information leakage, with the same model-selection procedure applied to both representations. An additional model using all shared genes will serve as a secondary gene-level baseline.

The final models will then be frozen and evaluated without retraining on GSE32280 and GSE39653. Cross-cohort performance will be compared primarily using AUROC, with F1-score, sensitivity, and specificity as complementary metrics. Better pathway-level performance would support improved cross-cohort generalization, while similar or lower performance would suggest little advantage from pathway aggregation or possible loss of gene-level predictive information.

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